Assignment 2

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Course: B.Tech ECE (1st Year)

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Task 1: User Persona

Product Context: A startup developing "ErgoSync," an AI-powered ergonomic desk chair cushion that monitors posture and provides real-time haptic feedback and app analytics.

Demographic Information:

- **Name:** Rohan Mehta
- **Age:** 28
- **Occupation:** Senior Software Developer (Remote Worker)
- **Location:** Tier-1 Metropolitan City
- **Income:** Upper-middle class

Goal and Objectives:

- Maintain a healthy posture during long, 8+ hour coding sessions.
- Prevent the recurrence of lower back pain without having to buy a completely new, expensive chair.
- Integrate health tracking seamlessly into his existing smart-device ecosystem.

Psychographic Information:

- **Interests:** Tech gadgets, fitness, home office setups, productivity optimization.
- **Personality Traits:** Analytical, health-conscious, introverted, data-driven.
- **Choices:** Prefers minimalist aesthetics, values functionality and data privacy over flashy marketing.

Behaviour and Preference:

- Spends prolonged periods seated and often forgets to take breaks.
- Relies heavily on wearable technology (smartwatches) for health metrics.
- Prefers non-intrusive notifications over loud alarms.

User Journey (Brief):

- Experiences back pain -> Searches online for ergonomic solutions -> Discovers

  ErgoSync cushion -> Reviews technical specs and buys -> Places on existing chair -> Uses the companion app to track daily posture score and receives gentle haptic nudges when slouching.

Challenges and Pain Points:

- Chronic lower back pain affecting productivity.
- Finds it difficult to consciously remember to sit straight when deep into complex coding tasks.
- Frustrated by traditional "ergonomic" products that lack quantitative data on whether they are actually working.

Task 2: User Journey Map (Samsung 2024 AI Family Hub Smart Fridge)

Scenario: A health-conscious user relying on the AI fridge to manage groceries, prevent food waste, and plan dietary-specific meals.

Stage	User Action	Fridge Feature / Touchpoint	User Emotion
1. Stocking Inventory	User returns from the grocery store and places items inside the fridge.	**AI Vision Inside:** Internal cameras automatically identify up to 33 fresh food items and add them to the digital inventory.	Impressed / Relieved (Saves time on manual entry).
2. Managing Expirations	User checks the fridge screen a few days later to see what needs to be eaten.	**Expiration Alerts:** The user manually inputs expiration dates on the 32-inch Family Hub screen. The fridge alerts the user about items nearing expiration.	Organized / Proactive (Prevents food waste).
3. Meal Planning	User is unsure what to cook for dinner with the remaining ingredients.	**Samsung Food App:** User accesses the app, which suggests recipes based on the existing internal inventory.	Inspired / Supported.
4. Personalization	User wants a meal that fits their specific diet (e.g., gluten-free, dairy-free).	**Vision AI Personalization:** The app filters recipe recommendations to accommodate the user's saved dietary restrictions.	Satisfied / Valued.
5. Health Integration	User aims to align their diet with their current fitness goals.	**Samsung Health Sync:** The fridge connects with the user's Samsung Health profile to recommend meals tailored to their specific health and fitness data.	Empowered (Holistic health management).

Task 3: Product Storyboard

Product: "FloraMind" - A smart IoT soil sensor and companion app for indoor plants.

- **Frame 1: The Problem**
  - ***Visual Description:*** A user looking disappointed at a wilted, yellowing indoor potted plant on their desk.
  - ***Caption:*** "Keeping indoor plants alive can be a guessing game."
- **Frame 2: The Setup**
  - ***Visual Description:*** The user unboxes a sleek, pen-like sensor (FloraMind) and inserts it directly into the plant's soil.
  - ***Caption:*** "Insert the FloraMind sensor into the soil of any plant."
- **Frame 3: The Connection**
  - ***Visual Description:*** The user's smartphone screen displays the FloraMind app, pairing via Bluetooth and selecting the plant species (e.g., 'Monstera').
  - ***Caption:*** "Sync with the app and select your plant species."
- **Frame 4: The Alert (Actionable Insight)**
  - ***Visual Description:*** A push notification appears on the phone: *"Your Monstera is thirsty! Moisture level at 15%."*
  - ***Caption:*** "Receive real-time alerts for water, sunlight, and fertilizer needs."
- **Frame 5: The Action**
  - ***Visual Description:*** The user waters the plant. An LED ring on the top of the physical soil sensor pulses green to indicate the water level is now optimal.
  - ***Caption:*** "Water perfectly with visual confirmation from the device."
- **Frame 6: The Result**
  - ***Visual Description:*** A few weeks later, the plant is vibrant and growing new leaves. The user smiles while drinking coffee next to it.
  - ***Caption:*** **"Enjoy thriving plants with zero guesswork."**

Task 4: Smart Trash Bins Scenario

Part A: Empathy Map

Target Demographic: City Pedestrians & Older Generation Citizens.

- **SAYS:**
    - "I don't know how to open this thing."
    - "Why is the light blinking red? Is it broken?"
    - "The old foot-pedal bins were much easier to use."
- **THINKS:**
    - "This technology is overly complicated for just throwing away trash."
    - "I don't want to touch the lid if the sensor is broken, it's unhygienic."
    - "The city wasted tax money on these."
- **DOES:**
    - Waves hand frantically over the sensor multiple times.
    - Leaves the trash bag on the ground next to the bin when it won't open.
    - Avoids the smart bins entirely and looks for a traditional bin.
- **FEELS:**
    - Frustrated by the inconsistent technology.
    - Confused by the unclear LED signals.
    - Disgusted by the overflowing trash around the malfunctioning bins.

Part B: Design Thinking Approach

1. Empathize: Observation reveals that users, particularly older generations, find the purely sensor-based interaction frustrating and unreliable. The lack of tactile control and confusing LED signals lead to abandonment and littering.

2. Define: The core problem is the lack of an intuitive, mechanical fallback mechanism for when sensors fail, compounded by a confusing user interface (LEDs) that fails to clearly communicate the bin's status (e.g., full, broken, or processing).

3. Ideate:

- *Solution 1:* Introduce a mechanical foot pedal as a secondary, fail-safe opening method. This bridges the gap for users who prefer traditional methods.
- *Solution 2:* Replace ambiguous colored LEDs with clear, universally understood backlit icons (e.g., a "Hand" icon for motion, a red "Trash Bag" icon for full).
- *Solution 3:* Recalibrate sensor placement to angle slightly outward, capturing user intent more reliably without requiring exact hand placement.
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4. Prototype: Develop a physical mock-up of the "Hybrid Smart Bin." It features the original sensor technology but now includes a sturdy stainless-steel foot pedal at the base. The LED strip is replaced by a small e-ink display or backlit panel showing simple symbols (🟢 "Ready", 🔴 "Bin Full").

5. Test: Deploy 20 prototyped "Hybrid Bins" in high-traffic public parks with a diverse age demographic. Measure success by monitoring the reduction of overflow alerts, observing the frequency of foot-pedal usage versus sensor usage, and conducting brief on-site interviews with older citizens to gauge improved satisfaction and adoption rates.